CHE 2201 – Physical Chemistry II

Quantum Mechanics

Lec.2. Basic Principles of Quantum Mechanics

2.1. Quantization of Energy

We have already discussed this aspect in section 1.2. It says that energy is not propagated and transmitted, absorbed or emitted continuously but discretely, in separate portions or quanta.

So, the energy E of an electromagnetic radiation of frequency υ are integer multiples of $h\upsilon$.

$$E = nh\upsilon \qquad (2.1),$$

where $n = 0, 1, 2, \ldots\ldots\ldots$; and h is the Planck's constant with the value 6.63×10^{-34} J s.

2.2. Wave-Particle Duality

In section 1.4, we have learned that the photoelectric effect could be explained by light (electromagnetic radiation) travelling as particles called photons with the energy of integer multiples of $h\upsilon$. On the other hand, most of the properties of light (such as diffraction and interference of light) can only be explained by assuming that light behaves as waves. However, at that time, not any scientist had taken the view that the matter is wave-like.

Nevertheless, experiments carried out in 1925, forced people to consider that possibility. The crucial experiment was performed by the American physicists Clinton Davisson and Lester Germer, who observed the diffraction of electrons by a crystal. Almost, at the same time, G.P. Thomson, working in Scotland, showed that a beam of electrons was diffracted when passed through a thin gold foil. The Davisson–Germer experiment, which has since been repeated with other particles (including α particles and molecular hydrogen), showed clearly that particles have wave-like properties.

Further, as mentioned earlier, we have also seen that waves of electromagnetic radiation have particle-like properties (e.g. photoelectric effect). Thus, we have now come to the heart of modern physics. When examined on an atomic scale (i.e. micro scale), the classical concepts of particle and wave diminishes. It means that the particles show the characteristics of waves, while the waves show the characteristics of particles.

So, we can conclude that, not only the electromagnetic radiation has the character classically attributed to particles, but electrons (and all other particles) have the characteristics classically attributed to waves. This joint particle and wave character of matter and radiation is called wave–particle duality. This concept is contradictory to one of the main principles of classical physics, where particles and waves are treated as entirely distinct entities.

In 1924, French physicist Louis de Broglie suggested extending this dual nature (i.e. particle-wave concepts) to all microparticles. It means that the motion of any microparticle be regarded as a wave process.

Mathematically this is expressed by the de Broglie equation, according to which a particle of mass m moving with a momentum p ($p = mv$, where v is the velocity of the particle) has a certain wavelength λ:

$$\lambda = h/p \qquad (2.2)$$

where h is the Planck's constant.

Equation 2.2 relates the properties of a particle (mass, velocity, momentum etc.) with the properties of a wave (wavelength). Further, it implies that a particle with a high momentum has a short wavelength. Macroscopic bodies have such high momentum (because their mass is so great), even when they are moving slowly, that their wavelengths are undetectably small, and the wave-like properties cannot be observed. This is why, we can use the classical mechanics to explain the behaviour of macroscopic bodies. However, when we are dealing with microscopic systems, such as atoms and molecules, in which masses are small, it is necessary to use quantum mechanics.

Example 1: Calculate the de Broglie wavelength of an electron moving with the velocity of 10^7 m s^{-1} (mass of the electron = 9.11×10^{-31} kg).

$\lambda = h/p = h/mv = (6.63 \times 10^{-34}$ J s) / (9.11×10^{-31} kg $\times$ 10^7 m s^{-1}) = 7.28×10^{-11} m.

Example 2: Calculate the de Broglie wavelength of a stone of mass 50 g, which was thrown with a velocity of 40 m s^{-1}.

$\lambda = h/p = h/mv = (6.63 \times 10^{-34}$ J s) / (50.0×10^{-3} kg $\times$ 40 m s^{-1}) = 3.32×10^{-34} m.

2.3. Uncertainty Principle

When we studied the atomic structure (under CHE 1201 – General Chemistry), it was stated that the classical model of the atom require precise information about the position of an electron and its momentum. So, suppose that we would like to measure the position and the momentum (or velocity) as accurately as possible.

In order to 'see' the electron and to determine its location, light (stream of particles called photons) should reflect from the electron, and pass through the microscope to our eyes. However, when a photon collides with an electron, it will transfer some of its energy to the electron, thus changing the electron's energy and its momentum. It means that while attempting to locate (to measure) the position of the electron, we have caused a change (uncertainty) in its momentum. This leads to the idea that the position and the momentum of an electron cannot be measured accurately at the same time, and it is true for any microparticle.

German physicist Werner Heisenberg put forward the principle (in 1927), that *"it is impossible to accurately determine both the position (coordinates) and the amount of motion (momentum = mv) of a microparticle simultaneously".*

The mathematical expression of the Uncertainty Principle is:

$$\Delta x\, \Delta p \geq h/2\pi \quad \text{or} \quad \Delta x\, \Delta v \geq h/2\pi m \qquad (2.3)$$

where Δx, Δp and Δv are the uncertainties of position, momentum and velocity of the particle respectively, h is the Planck's constant and m is the mass of the particle.

It follows from equation 2.3, that higher the accuracy of determination of the coordinate of a particle (i.e. the lower the uncertainty Δx), the less certain the values of momentum and velocity will be (i.e. higher the Δp and Δv), and vice versa. Thus, the more accurate the momentum (and velocity), the less certain the position of the particle. The ratio h/m is very small for macroparticles (heavy particles), so that laws of classical mechanics hold for them, and the position and velocity of a macro-body can be determined at the same time within their context.

For example, let us consider the ratio h/m for an electron.

$h = 6.63 \times 10^{-34}$ J s;

Mass of the electron = 9.11×10^{-31}kg

So, for an electron: $h/m = (6.63 \times 10^{-34})/(9.1\times 10^{-28}) \sim 10^{-6}$.

Now let us consider a macroparticle with the mass of 1 kg.

For such a macroparticle: $h/m = (6.63 \times 10^{-34})/1 \sim 10^{-34}$.

2.4. Schrödinger Equation

The quantization of energy, the wave–particle dual nature and the uncertainty principle, all indicate that the classical mechanics is unsuitable for describing the behaviour of microparticles. Hence, as indicated in the previous lecture, scientists had to construct a new mechanics – Quantum Mechanics.

Quantum mechanics acknowledges the wave–particle duality of matter and suppose that a particle is distributed through space like a wave, rather than travelling along a definite path. In other words, quantum mechanics replaces the classical concept of trajectory (path) of a particle by a wave, and its mathematical representation is called a Wavefunction (ψ).

In 1926, the Austrian physicist Erwin Schrödinger proposed an equation for finding the wavefunction of any system. The time-independent Schrödinger Equation for a particle of mass m moving in one dimension with energy E, in a system that does not change with time (for example, its volume remains constant) is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+V(x)\psi=E\psi \qquad (2.4),$$

where $V(x)$ is the potential energy of the particle at the point x;

the first term is related to the kinetic energy of the particle (the total energy E is the sum of potential and kinetic energies);

and $\hbar = h/2\pi$ (h-bar) is a convenient modification of Planck's constant with the value 1.055×10^{-34} J s.

The Schrödinger equation for a particle of mass m moving in a three-dimensional space is written as follows.

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi = E\psi \qquad (2.5)$$

Here, $\nabla^2\Psi = \partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 + \partial^2\psi/\partial z^2$ (2.6)

In quantum mechanics, the Schrödinger equation replaces the Newton's equation in classical mechanics ($F = m \times a$). In other words, the Schrödinger equation should be considered as the quantum mechanical postulate of motion that replaces the classical mechanical postulate of motion.

The credibility of the Schrödinger equation could be justified by showing that it implies de Broglie equation (equation 2.2) for a freely moving particle in a region, where its potential energy V is constant.

After writing V(x) = V, we can rearrange the equation 2.4 as follows.

$$\frac{d^2\psi}{dx^2}=-\frac{2m}{\hbar^2}(E-V)\psi \qquad (2.7)$$

A solution of the above differential equation could be written as

$\psi = \cos kx$, (2.8)

where

$$k = \left\{\frac{2m(E-V)}{\hbar^2}\right\}^{1/2} \qquad (2.9)$$

If we were to compare the equation 2.8 with the standard form of a harmonic wave, $\cos(2\pi x/\lambda)$, we can say that $\cos kx$ is a wave of wavelength $\lambda = 2\pi/k$.

On the other hand, the quantity (E – V) (i.e. total energy minus potential energy) is equal to the kinetic energy of the particle E_k.

From equation 2.9, it follows that

$$k = (2mE_k/\hbar^2)^{1/2} \qquad (2.10)$$

$$E_k = k^2\hbar^2/2m \qquad (2.11)$$

On the other hand, the kinetic energy of a particle

$$E_k = ½\ mv^2 = p^2/2m \qquad (2.12)$$

So, from equations 2.11 and 2.12, it follows that

$$p = k\hbar \qquad (2.13)$$

As such, the linear momentum is related to the wavelength of the wavefunction

$$p = \frac{2\pi}{\lambda} \times \frac{h}{2\pi} = \frac{h}{\lambda} \qquad (2.14)$$

Equation 2.14 represents the de Broglie equation (see equation 2.2).

2.5. Born Interpretation of the Wavefunction

A central postulate of quantum mechanics is that the wavefunction contains all the dynamical information about the system it describes. However, at this point, let us concentrate on the information it carries about the location of the particle.

The interpretation of the wavefunction in terms of the location of the particle is based on a suggestion made by German physicist and mathematician Max Born. He made use of an analogy with the wave theory of light, in which the square of the amplitude of an electromagnetic wave in a region is interpreted as its intensity, and therefore, in quantum terms, as a measure of the probability of finding a photon present in the region.

The Born interpretation of the wavefunction focuses on the square of the wavefunction, i.e. $|\psi|^2 = \psi^*\psi$, where ψ^* is the complex conjugate of ψ.

So, for a particle moving on one-dimension, the Born interpretation can be formulated as follows.

If the wavefunction of a particle has the value ψ at some point x, then the probability of finding the particle between x and $(x+dx)$ is proportional to $|\psi|^2 dx$. Thus, $|\psi|^2$ is the probability density, and to obtain the probability it must be multiplied by the length of the infinitesimal region dx. The wavefunction ψ itself is called the probability amplitude.

For a particle moving in three dimensional space (for example, an electron near a nucleus in an atom), the wavefunction depends on the point r with coordinates x, y and z, and the interpretation of $\psi(r)$ is as follows (see Fig. 2.1).

If the wavefunction of a particle has the value ψ at some point r, then the probability of finding the particle in an infinitesimal volume $d\tau = dxdydz$ at that point is proportional to $|\psi|^2 d\tau$.

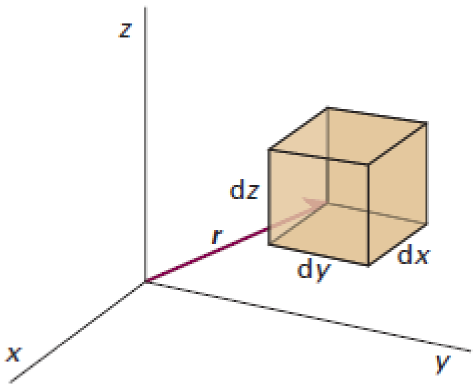

Fig. 2.1. Born Interpretation of the Wavefunction 3-D Space

The Born interpretation does away with any worry about the significance of a negative (and, in general, complex) value of ψ, because $|\psi|^2$ is real and never negative. It means that there is no direct significance in the negative (or complex) values of a wavefunction, as only its square, which is a positive quantity, is directly physically significant. However, both the negative and positive regions of a wavefunction may correspond to a high probability of finding a particle in a region.

2.5.1. Normalization

A mathematical feature of the Schrödinger equation is that, if ψ is a solution, and then $N\psi$ is also a solution, where N is any constant. This feature is confirmed by noting that ψ occurs in every term in equation 2.4, so any constant factor can be cancelled. This freedom to vary the wavefunction by a constant factor means that it is always possible to find a <u>normalization constant</u>, N, such that the proportionality of the Born interpretation becomes an equality.

[In simple terms, normalization is a statement that the particle exists at some point at all times. In other words, if the probability is equal to zero, then it means the particle would not exist.]

We can find the normalization constant by assuming that, for a normalized wavefunction $N\psi$, the probability that a particle is in the region dx is equal to $(N\psi^*)(N\psi)\mathrm{d}x$. Furthermore, the sum over all space of these individual probabilities must be 1, as the probability of the particle being somewhere is 1.

This requirement can be expressed mathematically as follows.

$$N^2 \int_{-\infty}^{+\infty} \psi^* \psi dx = 1 \qquad (2.15)$$

From now on, unless it is stated otherwise, we will use wavefunctions that have been normalized to 1.

That is, we assume that ψ already includes a factor that ensures that (in one dimension)

$$\int_{-\infty}^{+\infty} \psi^* \psi dx = 1 \qquad (2.16)$$

In three-dimensions, the wavefunction is normalized, if

$$\int_{-\infty}^{+\infty} \psi^* \psi d\tau = 1 \qquad (2.17),$$

where $d\tau = dxdydz$.

2.5.2. Acceptability of Wavefunctions

The Born interpretation puts several restrictions on the acceptability of wavefunctions. The principal requirement is that ψ must not be infinite anywhere. If it is infinite, it means that the integral in equations 2.16 and 2.17 would be infinite (in other words, ψ would not be square-integrable) and the normalization constant would be zero. The normalized function would then be zero everywhere, except where it is infinite, which would be unacceptable. The requirement that ψ is finite everywhere rules out many possible solutions of the Schrödinger equation, because many mathematically acceptable solutions rise to infinity, and are therefore physically unacceptable.

The requirement that ψ is finite everywhere is not the only restriction implied by the Born interpretation. We could imagine a solution of the Schrödinger equation that gives rise to more than one value of $/\psi/^2$ at a particular single point. The Born interpretation implies that such solutions are unacceptable, because it would be absurd to have more than one probability that a particle is at the same point. This restriction is expressed by saying that the wavefunction must be single valued, that is, the wavefunction must have only one value at each point of space.

The Schrödinger equation itself also implies some mathematical restrictions on the type of functions that are acceptable (i.e. physically possible). Because it is a differential equation of second-order, the second derivative of ψ must be well-defined if the equation is to be applicable everywhere. We can take the second derivative of a function only if it is continuous, and if its first derivative, its slope, is also continuous.

Thus, we can conclude that the acceptable solutions to the Schrödinger equation (i.e. solutions which are physically possible), must have certain properties.

Accordingly, the possible wavefunction ψ must

- be continuous;
- have a continuous slope;
- be single-valued;
- be square-integrable.

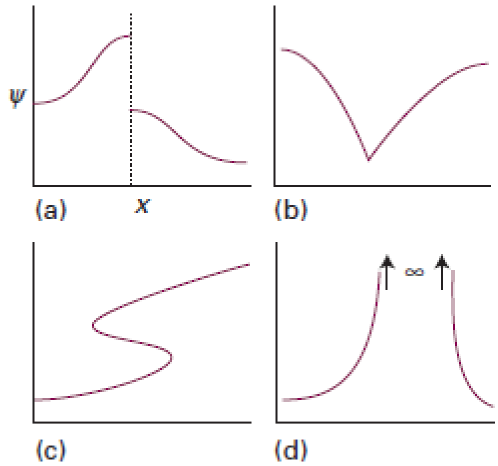

Fig. 2.2. Unacceptable Solutions of the Schrödinger Equation

(a). Unacceptable because it is Not Continuous; (b). Unacceptable because its' Slope is Discontinuous; (c). Unacceptable because it is Not Single-Valued; (d). Unacceptable because it is Infinite

These are such severe restrictions that acceptable solutions of the Schrödinger equation do not in general exist for arbitrary values of the energy E. In other words, a particle may possess only certain energies, as otherwise its wavefunction ψ would be physically unacceptable. That is, as a consequence of the restrictions on its wavefunction, the energy of a particle is quantized. In other words, the quantization of energy of a particle flows from the Schrödinger equation itself, and from the physically acceptable wavefunctions. We can find the acceptable energies by solving the Schrödinger equation for motion of various kinds, and selecting the solutions that conform to the restrictions discussed above.